1 Will and Olly share £80 in the ratio 3 : 2

Work out how much each of them get.

(3 marks)

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2 Molly, Paige and Demi share 42 sweets in the ratio 3 : 2 : 1

Work out the number of sweets that each of them receives.

(3 marks)

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3 *ABC* is a straight line.



The length of *BC* is three times the length of *AB*.

AC = 80 metres.
Work out the length *BC*.

(3 marks)

---

4 Carly and James share some money in the ratio 5 : 3
Carly gets £70 more than James.

Work out how much money James gets.

(3 marks)

---

5 Jerry and Mick share some money in the ratio 2 : 3
Mick gets £900

Work out how much money Jerry gets.

(3 marks)

---

6 Ali and Steve share some sweets in the ratio 2 : 7
Steve gets 30 more sweets than Ali.

Work out how many sweets Steve gets.

(3 marks)

---

7 Dave is making cookies.
He mixes flour, butter and sugar in the ratio 6 : 4 : 1

Dave uses 160 grams of butter.
Work out how much flour and sugar Dave uses.

(3 marks)

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8 Alvin and Simon shared £540 in the ratio 4 : 5

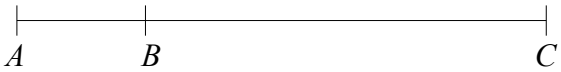
Alvin gave half of his share to Theo.
Simon gave a tenth of his share to Theo.

What fraction of the £540 did Theo receive?

(3 marks)

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9 *ABC* is a straight line.



The length of *BC* is four times the length of *AB*.

BC = 100 metres.
Work out the length *AC*.

(3 marks)

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10 Bob is going to make some orange paint. He needs to mix red paint, yellow paint and white paint in the ratio 5 : 4 : 1

Bob wants to make 750 m*l* of orange paint.
Bob has: 400 m*l* of red paint
300 m*l* of yellow paint
200 m*l* of white paint

Does Bob have enough red paint, yellow paint and white paint to make the orange paint?
You must show all your working.

(4 marks)

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11 Megan is going to make a drink using the instructions below.

Mix 2 parts of fruit juice
with 5 parts of sparkling water

Megan has 180 m*l* of fruit juice and 400 m*l* of sparkling water.

What is the greatest amount of the drink Megan can make?

(3 marks)

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12 In a bag there are only red counters, blue counters and white counters.
A counter is taken at random from the bag.

The table shows the probability of getting a red counter.

Colour	Red	Blue	White
Probability	0.35		

the number of blue counters : the number of white counters = 2 : 3

Find the probability of getting a blue counter and a white counter.

(4 marks)

---

13 Al, Tom and Joe share €3000.

The ratio of the amount Al gets to the amount Tom gets is in the ratio 5 : 4
Joe gets 1.5 times the amount Tom gets.

Work out the amount of money that Tom gets.

(4 marks)

---

14 Harry and Gary have a total of 300 stickers.
The ratio of the number of stickers Harry has to the ratio of the number of stickers Gary has is in the ratio 7 : 3

Harry gives Gary some stickers.
The ratio of the number of stickers Harry has to the ratio of the number of stickers Gary has is now in the ratio 8 : 7

Work out how many stickers Harry gives to Gary.
You must show all your working.

(4 marks)

---

15 A shop sells small chocolate bars and large chocolate bars.

There are
small chocolate bars are sold in packs of 4
large chocolate bars are sold in packs of 9

On one day

the number of packs of small chocolate bars sold : the number of packs of large chocolate bars sold = 5 : 2

A total of 190 chocolate bars were sold.
Work out the number of small chocolate bars sold.

(4 marks)

16 Dermot has 240 counters.
The counters are either red, or blue, or yellow or green.

15% of the counters are red.

$\frac{2}{5}$ of the counters are blue

The ratio of yellow counters to green counters is 3:1

Work out the number of yellow counters Dermot has.

(4 marks)

17 Daisy bakes 180 cakes.
She bakes only chocolate cakes, vanilla cakes, banana cakes and lemon cakes.

20% of the cakes are chocolate cakes.

$\frac{4}{9}$ of the cakes are vanilla cakes.

The ratio of banana cakes to lemon cakes is is 3:5

Work out the number of lemon cakes Daisy has.

(4 marks)

18 Each month Angelina earns £1800 after tax.

Angelina saves 30% of her salary and she spends the rest.

the amount Angelina spends on rent : the amount she spends on other expenses = 4 : 5

Work out how much Angelina spends on rent.

(4 marks)

19 There are only red counters and green counters in a bag.

number of red counters : number of green counters = 2 : 3

There are 28 red counters in the bag.

Work out the total number of counters in the bag.

(2 marks)

20 Clara and Dawn have some stamps in the ratio 5 : 9
Dawn has 36 more stamps than Clara.

Work out the number of stamps Clara has.

(3 marks)

21 In a bag there are counters.
The counters are all either red or blue or yellow.

The number of red counters	:	The number of blue counters	:	The number of yellow counters	=	4 : 5 : 8

The number of yellow counters is 24 more than the numbers of blue counters.

Work out the total number of counters in the bag.

(3 marks)

22 Cameron has some coins with a total value of 95 pence.

She has only 2 pence coins and 5 pence coins.
The ratio
number of 2 pence coins : number of 5 pence coins = 2 : 3

Work out how many 5 pence coins Cameron has.

(4 marks)

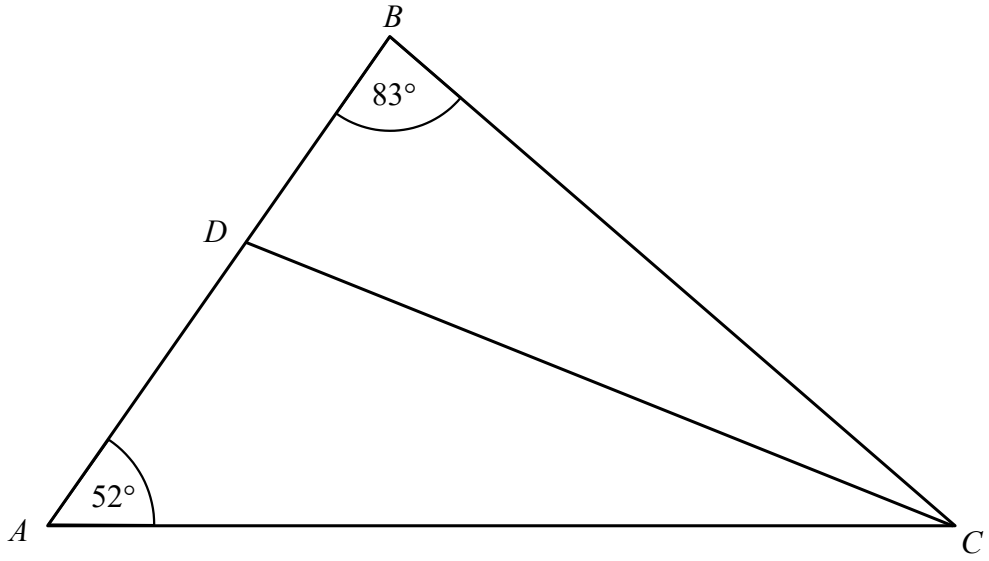
23 Pens and pencils are sold in a shop.

8 pencils cost £1.92
The ratio of the cost of a pen to the cost of a pencil is 4 : 3

Work out the cost of 5 pens.

(4 marks)

24 The diagram shows triangle *ABC*.



ADB is a straight line.

the size of angle *BCD* : the size of angle ACD = 2 : 3

Work out the size of angle *ADC*.

(4 marks)